Erasmus Mundus Joint Master Artificial Intelligence for Sustainable Societies (AISS)

Topics on Machine Learning and its Applications

Pedro Strecht

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Education

- PhD in Informatics Engineering (Machine Learning), Universidade do Porto
- MSc in Informatics and Computing Engineering (Requirements Engineering), Universidade do Porto
- BSc in Electrical and Computer Engineering (Telecommunications, Electronics and Computers), Universidade do Porto

Universidade Lusófona

- Invited Assistant Professor, Universidade Lusófona do Porto
- Teaching in Informatics Engineering Bacherol
- Teaching in Erasmus Mundus Joint Master - Artificial Intelligence for Sustainable Societies (AISS)

Universidade do Porto

- Head of Data, Architectures and Interoperability, Universidade do Porto Digital
- External Research Collaborator INESC TEC
- Laboratory of Artificial Intelligence and Decision Support (LIAAD)

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Learning Objectives

Data Science Fundamentals Fundamentals

- Identify key topics in data science project activities
- Main techniques of exploratory data analysis
- Main repositories of machine learning and datasets

Modeling & Evaluation

- Modelling and evaluation pipeline of predictive models
- Main techniques of data preprocessing
- Study the classic algorithms of classification and regression

Advanced Techniques

- Ensemble methods approaches
- Main clustering methods
- Methods of frequent pattern discovery

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Syllabus

01. Topics on ML and Applications

02. Introduction to Knowledge Discovery

03. Data representation

04. Repositories in machine learning

05. Exploratory data analysis

06. Model training

07. Model evaluation

08. Feature engineering

09. Classification: White-box models

10. Classification: Black-box models

11. Clustering methods

12. Frequent pattern discovery

13. Time Series Analysis

14. Text mining and NLP

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Students' evaluation

Written Tests (50%)

- 1st written test: **25/03/2026** (25%)
- 2nd written test: **20/05/2026** (25%)

Paper Discussion (10%)

- Presentation on a specific topic
- During the semester

Data Science Project (40%)

- Project with presentation
- Due: **27/05/2026**

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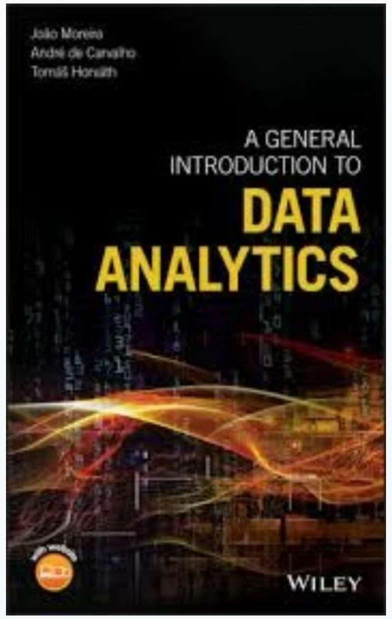
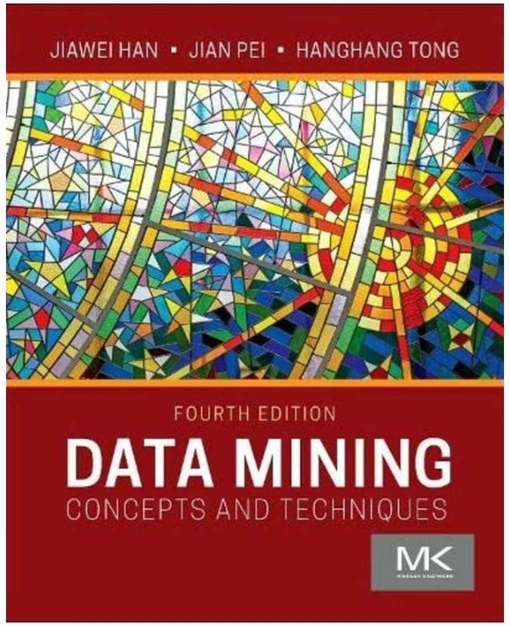
Main references



A General Introduction to Data Analytics

Moreira, J., Carvalho, A. & Horvath, T. (2019). Wiley



Data Mining: Concepts and Techniques (4th Edition)

Han, J., Pei, J. & Tong, H. (2022). Morgan Kaufmann

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Student introductions

Please introduce yourself

What is your name?

Your background

Where do you come from?

Your expectations

What do you expect from this course?

Your ML experience

On a scale of 0 (beginner) to 4 (expert), how would you rate yourself in machine learning?